INNOVATIVE MEDICAL DEVICE LABORATORY

Optimization of Electrode Configuration for Bladder-Targeted Temporal Interference Stimulation Using a Three-Dimensional Rat Model

Presenter | Thar Htet Nyan Principal Investigator | Eunkyoung Park

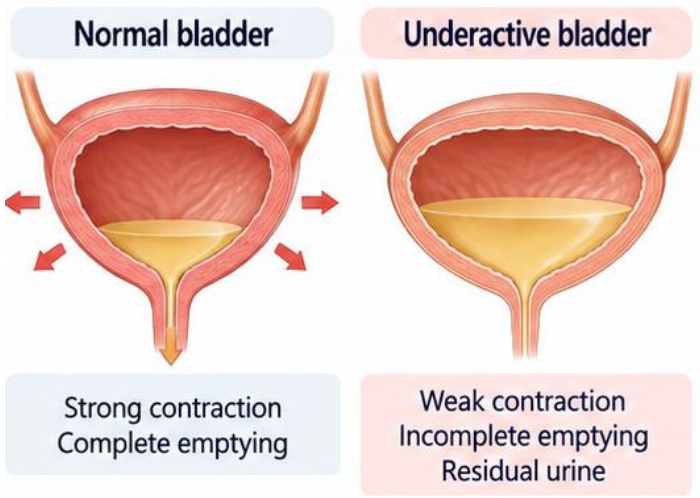
SCH Department of Biomedical Engineering, Soonchunhyang University, Asan, Republic of Korea

Background & Research Objectives

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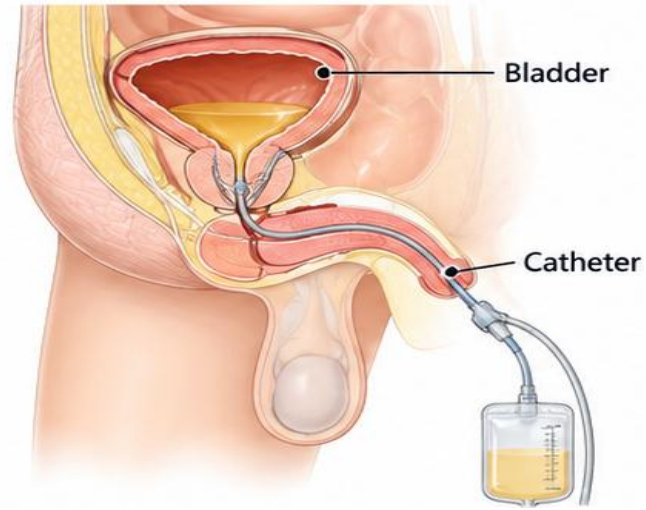
Background

Underactive Bladder (UAB)



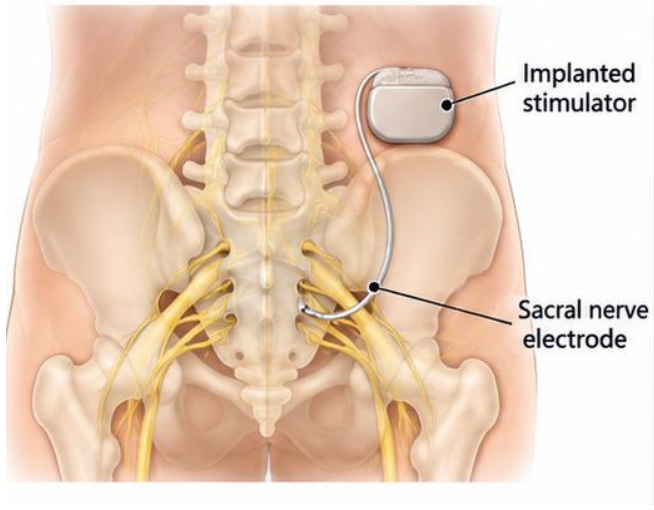
- Insufficient detrusor contractility resulting in incomplete bladder emptying.
- May lead to urinary retention and urinary tract complications.

Catheterization



- Requires repeated catheter insertion (intermittent or indwelling).
- Risk of urinary tract infections and patient discomfort.

Sacral Neuromodulation (SNM)



- Requires surgical implantation of a stimulation device.
- Involves long-term device management and potential surgical revisions.

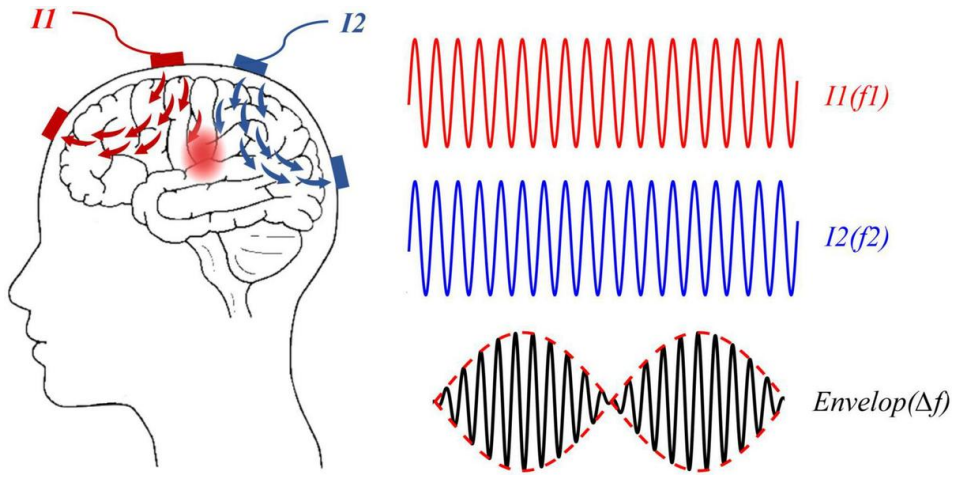
Research Need

A non-invasive neuromodulation approach capable of targeting bladder-related structures is needed.

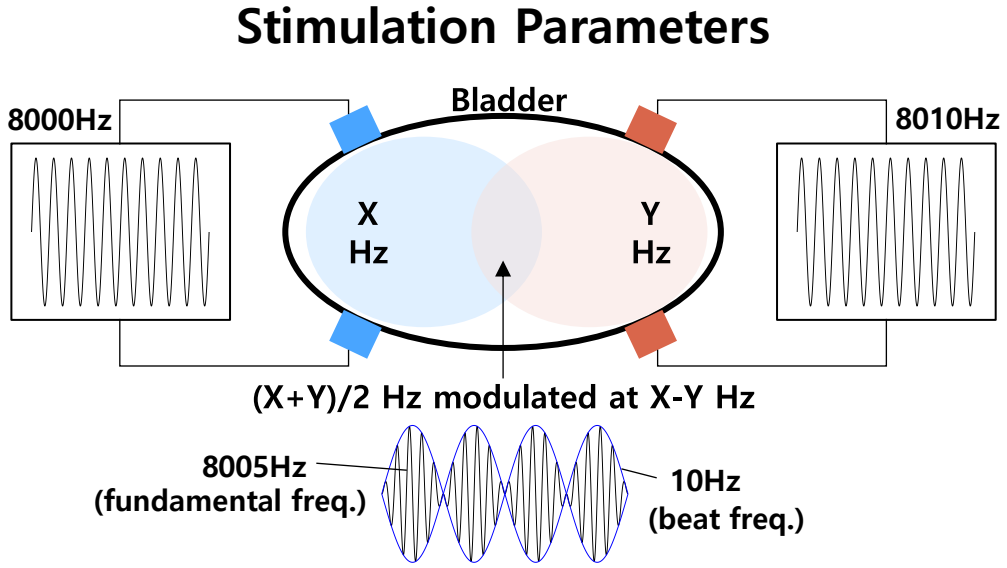
Temporal Interference Stimulation

Principle of Temporal Interference Stimulation (TIS)

- Temporal Interference Stimulation (TIS) is a non-invasive electrical neuromodulation technique designed to target deep neural structures.
- It uses two slightly different high-frequency electric fields to generate a low-frequency amplitude-modulation envelope.
- Primarily investigated for deep brain stimulation, TIS is also being explored for spinal cord and peripheral neuromodulation applications.



Deep Brain Stimulation



1. Guo W, He Y, Zhang W, Sun Y, Wang J, Liu S and Ming D (2023) A novel non-invasive brain stimulation technique: "Temporally interfering electrical stimulation". Front. Neurosci. 17:1092539. doi: 10.3389/fnins.2023.1092539

Research Objectives & Workflow

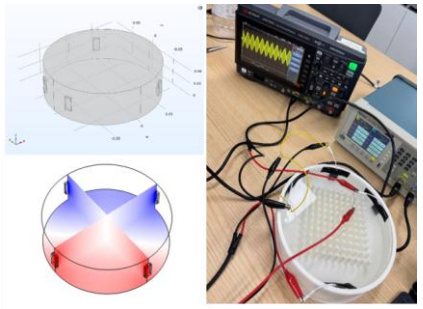
Main Objective

- To optimize electrode placement for bladder-targeted temporal interference stimulation using computational modeling and experimental validation.

Overall Workflow

I. Phantom

Experimental validation



II. 2D Rat Model

Preliminary placement



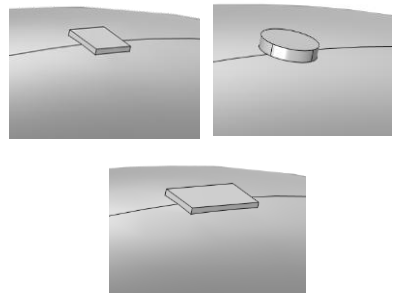
III. 3D Rat Model

3D Optimization



IV. Geometry

Electrode shape comparison

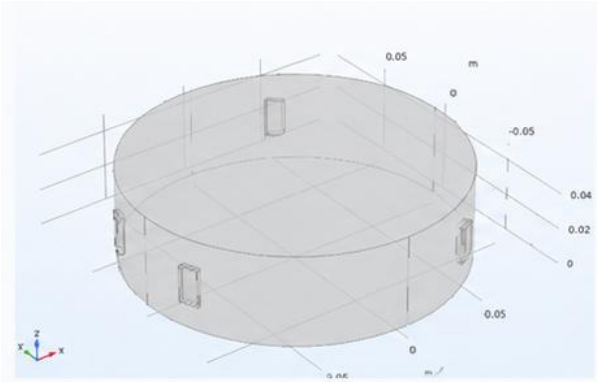


Phantom Experimental Validation

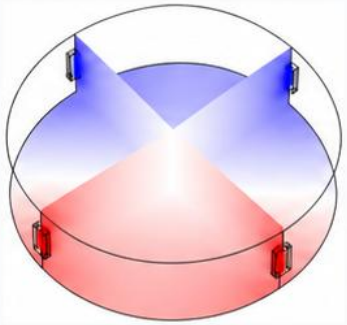
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Phantom Experimental Validation

COMSOL Stimulation



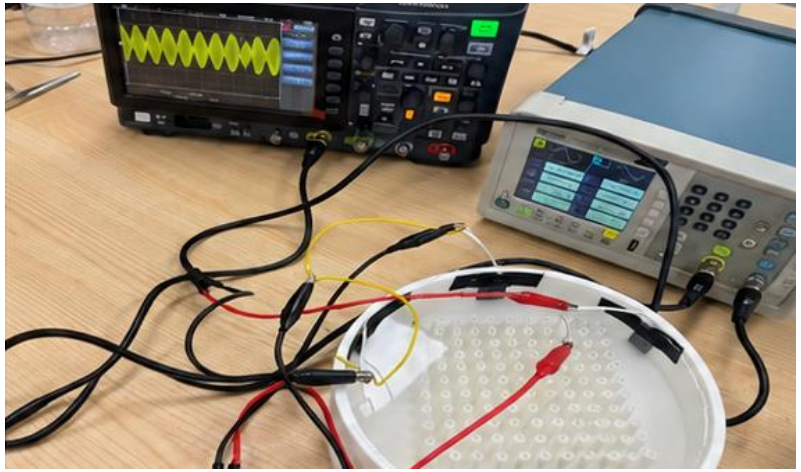
Cylindrical model with surface electrodes



Simulated electric-field distribution (top view)

- COMSOL Multiphysics 6.3
- Cylinder model: R = 95 mm, H = 50 mm
- Saline conductivity: 1.7 S/m; gel: 1.0 S/m
- Air treated as an insulator

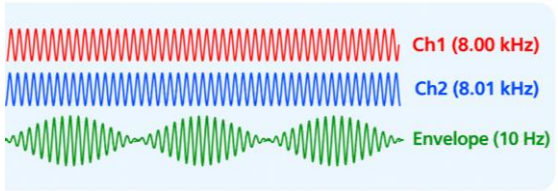
Physical Saline Phantom



- 3D-printed PLA circular container
- 0.9% NaCl saline, σ ≈ 1.7 S/m
- Surface electrodes: 1 × 2 cm
- 11 × 11 measurement grid (10 mm spacing)
- Six electrode configurations tested

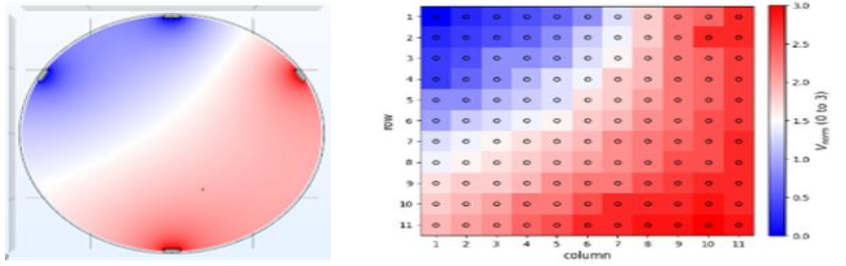
Stimulation Parameter

Channel 1	Channel 1	Frequency Difference (Envelope)
8.00 kHz (3.1 Vpp)	8.01 kHz (3.1 Vpp)	10Hz

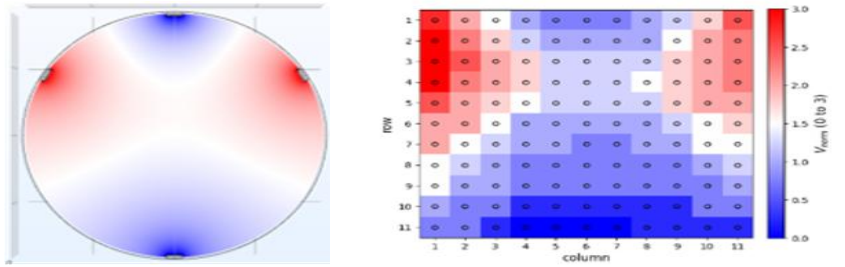


Phantom Experimental Validation: Result

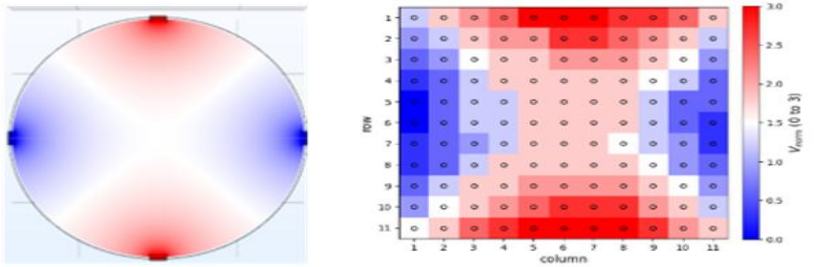
Case 1 COMSOL (Simulation) Experiment (Phantom)



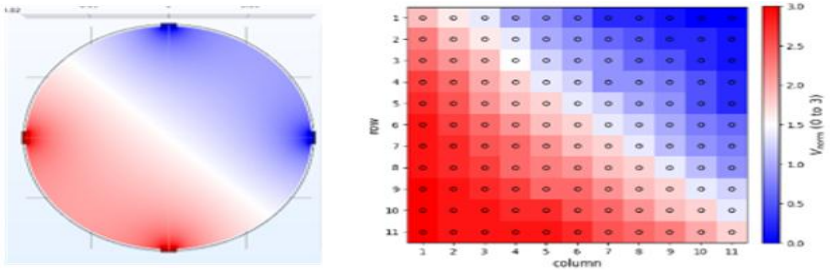
Case 2 COMSOL (Simulation) Experiment (Phantom)



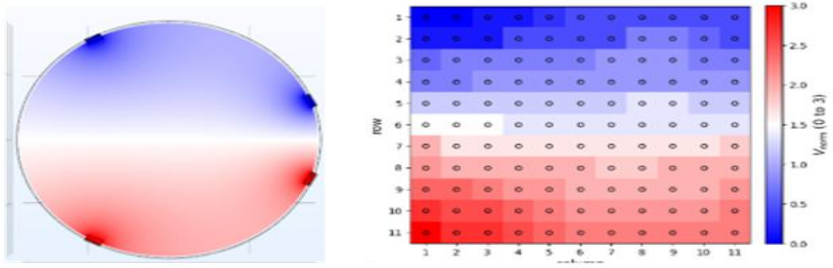
Case 3 COMSOL (Simulation) Experiment (Phantom)



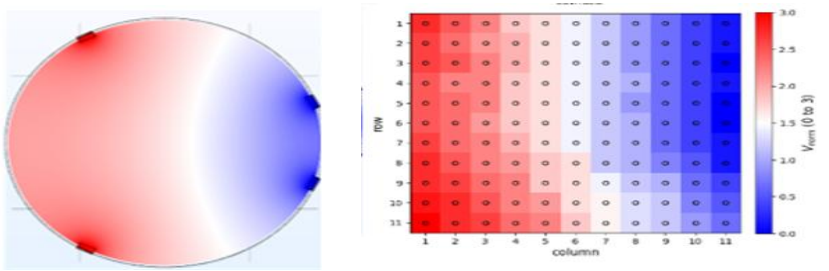
Case 4 COMSOL (Simulation) Experiment (Phantom)



Case 5 COMSOL (Simulation) Experiment (Phantom)



Case 6 COMSOL (Simulation) Experiment (Phantom)



Experimental measurements showed comparable spatial field-distribution trends to the COMSOL simulations.

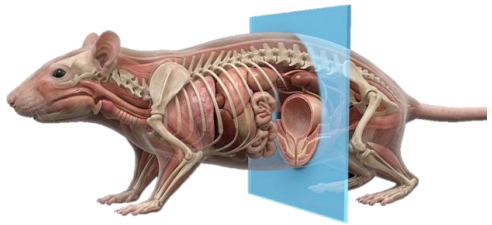
2D Rat Models

Preliminary Placement

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DEVICE
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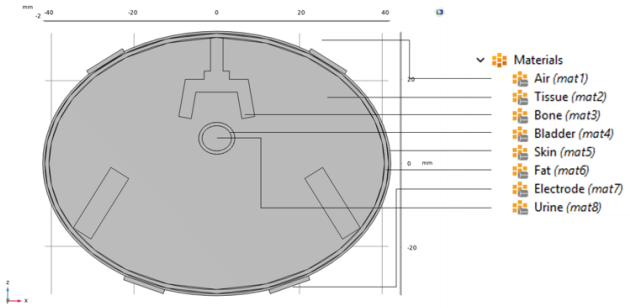
2D Rat Models: Stimulation Setup

Transverse Model (Cross-sectional)



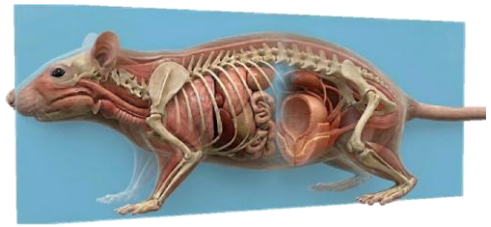
40 × 30 mm
2D model size

COMSOL Geometry



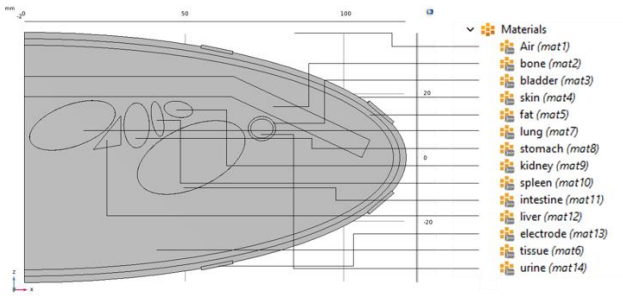
- Bladder-centered cross-section
- Model size: 40 × 30 mm
- Surface electrode placement
- 2D electric-field analysis

Sagittal Model (Longitudinal)



230 × 70 mm
2D model size

COMSOL Geometry



- Bladder-containing longitudinal section
- Model size: 230 × 70 mm
- Surface electrode placement
- 2D electric-field analysis

Material Properties (Electrical Conductivity)

Transverse model

- Tissue: σ = 0.339 S/m
- Bone: σ = 0.0204 S/m
- Bladder: σ = 0.2125 S/m
- Skin: σ = 0.00182 S/m
- Fat: σ = 0.0429 S/m
- Urine: σ = 1.5 S/m
- Electrode gel: σ = 1.0 S/m
- Air: σ = 1×10^{-14} S/m (insulator)

Sagittal model

Same as transverse model, plus:

- Lung: σ = 0.0919 S/m
- Stomach: σ = 0.529 S/m
- Kidney: σ = 0.339 S/m
- Spleen: σ = 0.1100 S/m
- Intestine: σ = 0.557 S/m
- Liver: σ = 0.0514 S/m

Stimulation Parameters

f_1 = 8,000 Hz **f_2 = 8,010 Hz**

Current = 3 mA

Frequency difference (envelope) = 10 Hz

Evaluation Metrics

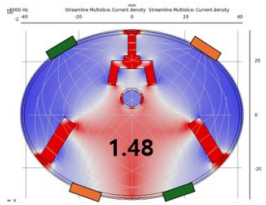
Selectivity: $\frac{E_{bladder}}{E_{tissue}}$

- Electric-field magnitude (V/m)
- Temporal interference (TI) envelope

2D Transverse Model: Electrode Placement Optimization

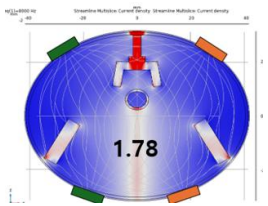
Nine Electrode Configurations (Transverse Section)

Case 1



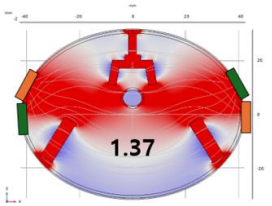
Selectivity = 1.48

Case 2



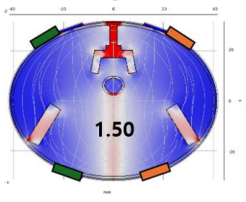
Selectivity = 1.78

Case 3



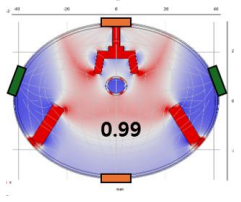
Selectivity = 1.37

Case 4



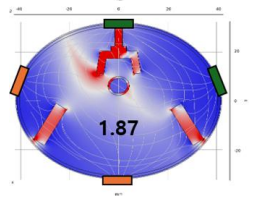
Selectivity = 1.50

Case 5



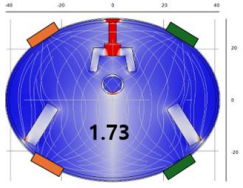
Selectivity = 0.99

Case 6



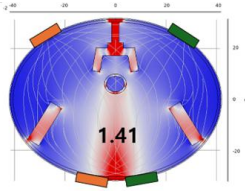
Selectivity = **1.87**

Case 7



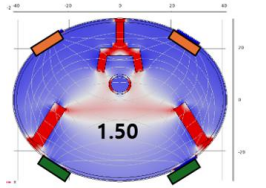
Selectivity = 1.73

Case 8

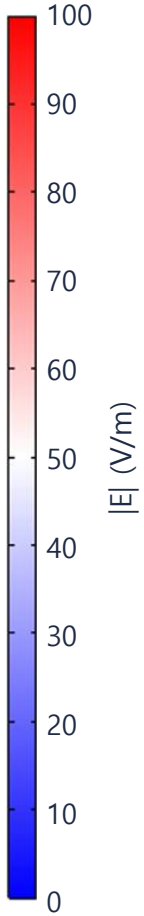


Selectivity = 1.41

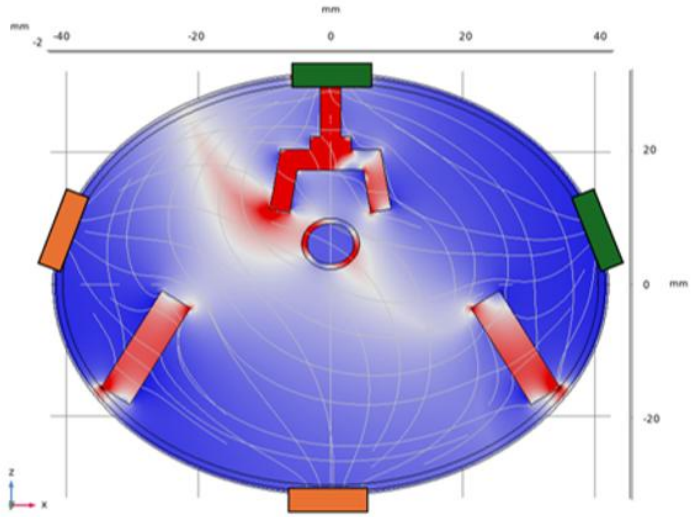
Case 9



Selectivity = 1.50



Optimal Configuration: Case 6



8000 Hz 8010 Hz

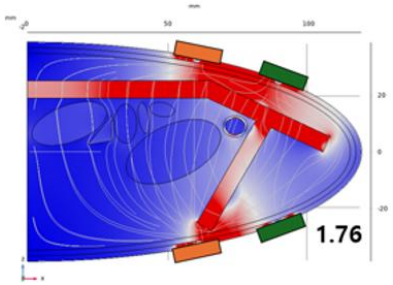
Mean bladder field (V/m)	**40.11**
Mean tissue field (V/m)	**21.40**
Bladder-to-tissue selectivity	**1.87**
Electrode arrangement	**Top–Diagonal**

Case 6 achieved the highest bladder-to-tissue selectivity among nine transverse electrode configurations.

2D Sagittal Model: Electrode Placement Optimization

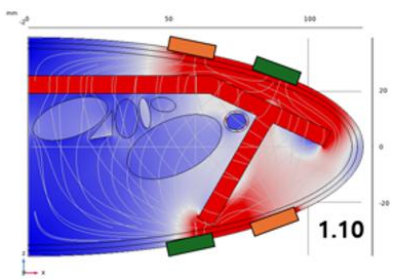
Five Electrode Configurations (Sagittal Section)

Case 1



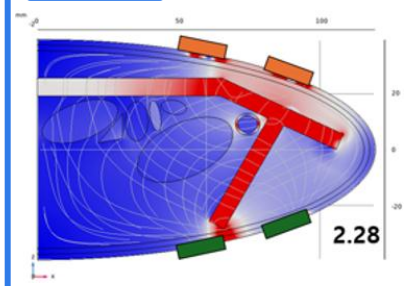
Selectivity = 1.76

Case 2



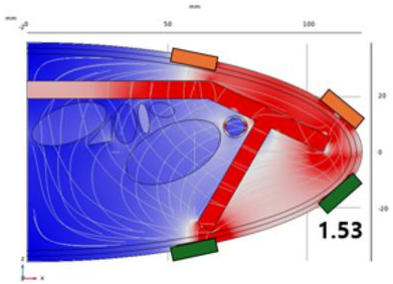
Selectivity = 1.10

Case 3



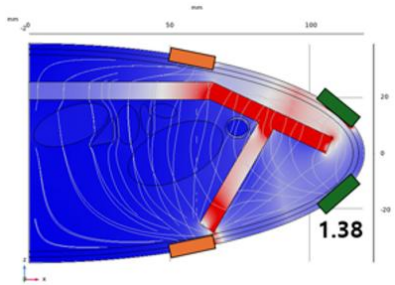
Selectivity = **2.28**

Case 4

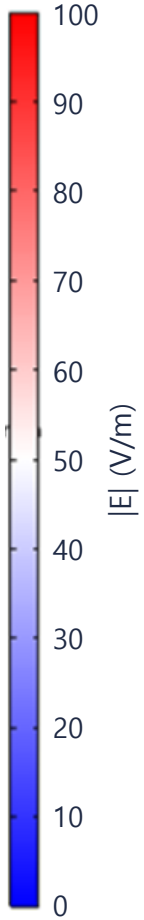


Selectivity = 1.53

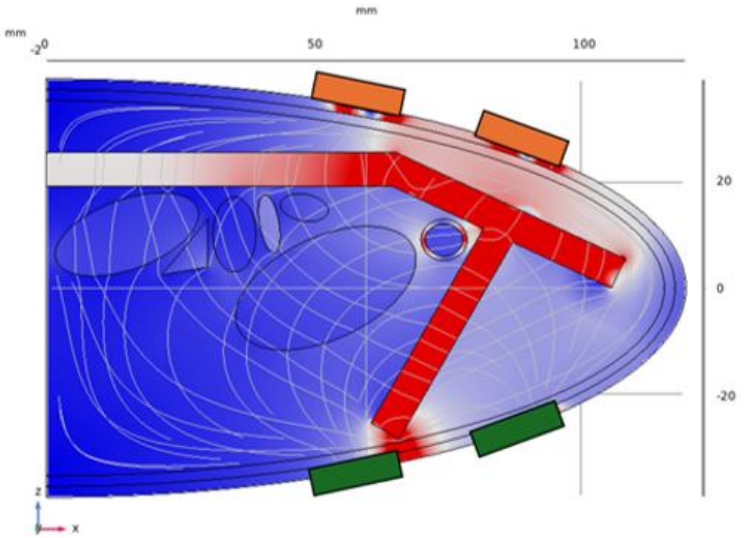
Case 5



Selectivity = 1.38



Optimal Configuration: Case 3



8000 Hz 8010 Hz

Mean bladder field (V/m)	**42.85**
Mean tissue field (V/m)	**18.79**
Bladder-to-tissue selectivity	**2.28**
Electrode arrangement	**Top–Diagonal**

Case 3 achieved the highest bladder-to-tissue selectivity among five sagittal electrode configurations.

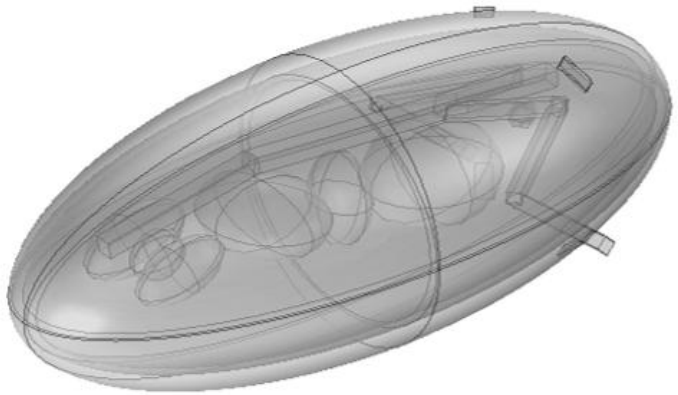
3D Rat Models Optimization



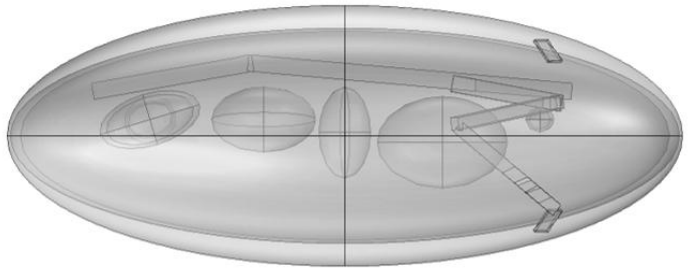
3D Rat Model: Simulation Setup

3D Anatomical Model

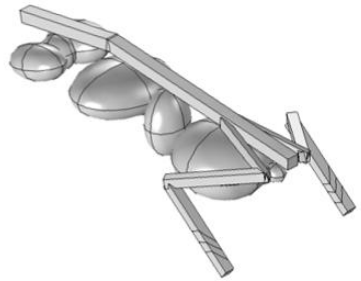
Oblique view



Orthogonal view



Structures



Model Setup and Simulation Parameters

Model Setup

Model dimensions	104 × 44 × 40 mm
Anatomical tissue structures	Included
Surface electrode placement	Five configurations compared
Analysis	3D electric-field analysis

Stimulation Parameters

Stimulation frequencies	f_1 = 8,000 Hz | f_2 = 8,010 Hz (Δf = 10 Hz)
Stimulation current	3 mA
Electrode configurations	5 configurations (surface electrodes)

Evaluation Metrics

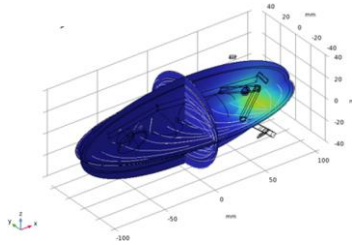
Selectivity	$S_{B/T} = \frac{\bar{E}_{bladder}}{\bar{E}_{tissue}}$
Electric-field magnitude	(V/m)
TI envelope	Temporal interference (TI) envelope

Objective: Optimize electrode placement to maximize the electric field in the bladder while minimizing off-target tissue exposure.

3D Electrode Configuration Comparison

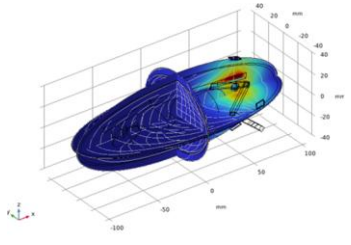
Five Surface-Electrode Configurations (3D Model)

Case 1



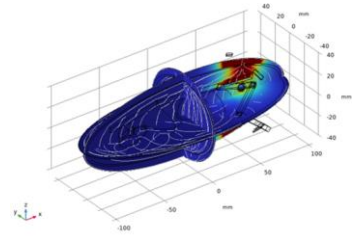
Selectivity = 6.83

Case 2



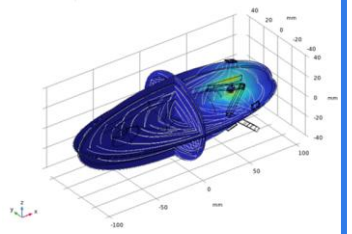
Selectivity = 5.47

Case 3



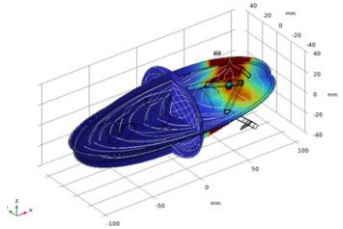
Selectivity = 6.20

Case 4

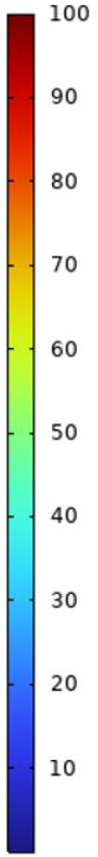


Selectivity = **7.96**

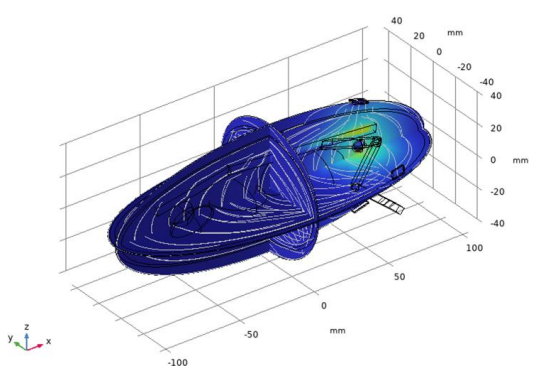
Case 5

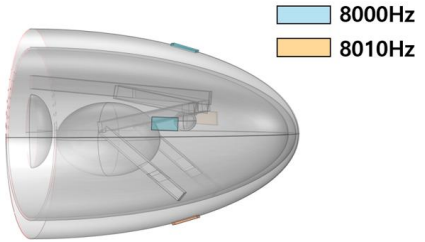


Selectivity = 5.12



Optimal Configuration: Case 4





Bladder/Tissue Selectivity



Quantitative Results

Mean bladder field	**63.30 V/m**
Mean surrounding-tissue field	**7.95 V/m**
Bladder-to-tissue selectivity	**7.96**

Case 4 achieved the highest bladder-to-tissue selectivity among the five 3D electrode configurations.

Optimal Electrode Placement Strategy

Stimulation Channels

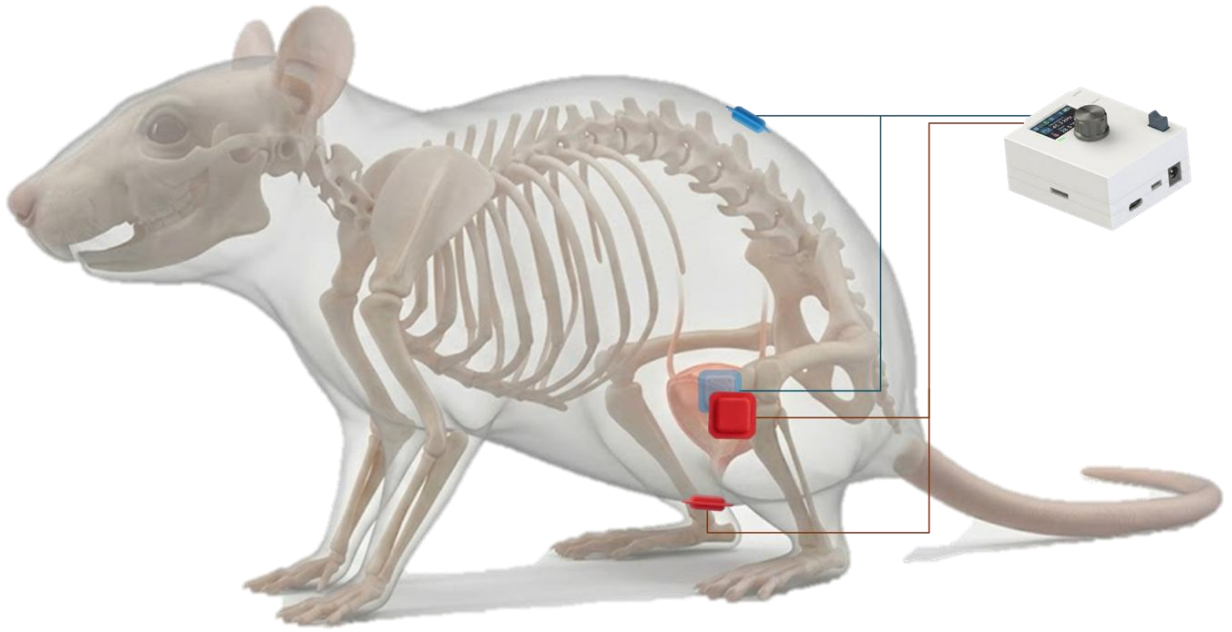
Channel 1 (Top–Side Pair)

Electrode	Upper (above bladder)
Return	Lateral (side)

Channel 2 (Bottom–Side Pair)

Electrode	Lower (below bladder)
Return	Opposite lateral side

Optimal Electrode Placement



Channel 1 – Top–Side Pair Channel 2 – Bottom–Side Pair

Two electrode pairs: Channel 1 (Top–Side) uses an upper electrode; Channel 2 (Bottom–Side) uses a lower one.

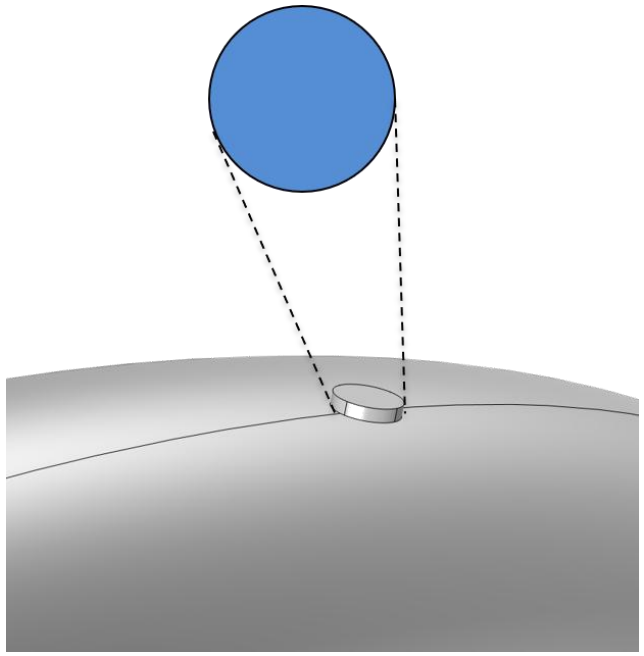
Electrode Geometry Comparison

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Electrode Geometry Comparison

Circle

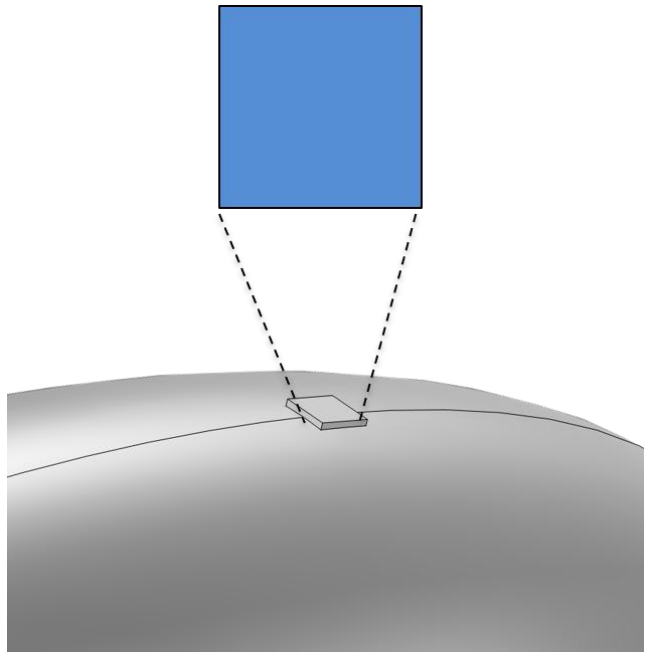
Radius: 2.5 mm



Selectivity **6.824**

Square

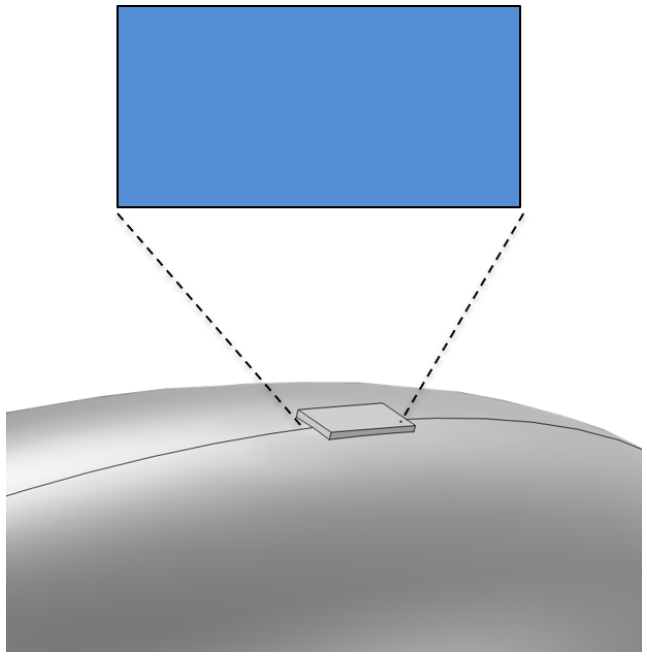
Length × Width: 3 × 3 mm



Selectivity **6.830**

Rectangle

Length × Width: 3 × 6 mm



Selectivity **6.828**

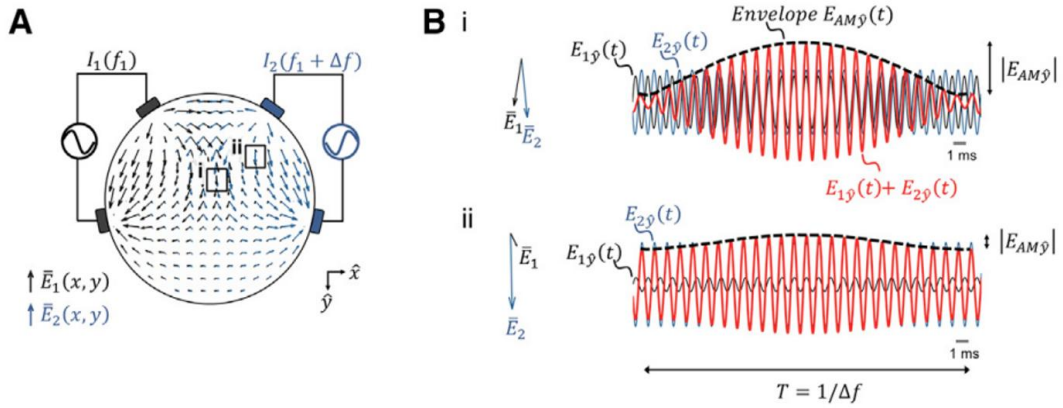
Only minor differences among the three geometries; the 3 × 3 mm square gave the highest selectivity (6.830).

Limitations and Future Work



Limitations and Future Work

Direction-dependent TI Envelope

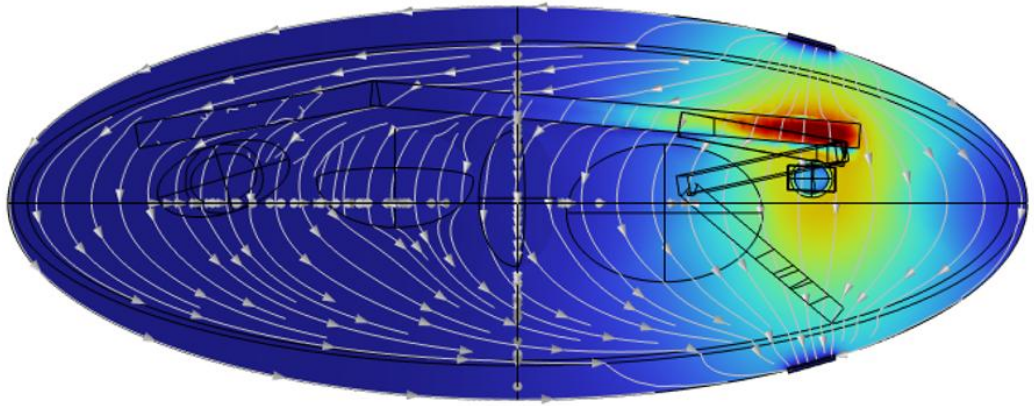


$$\Delta E(\hat{n}) = 2 \min(|E_1 \cdot \hat{n}|, |E_2 \cdot \hat{n}|)$$

Same location ≠ same effective modulation in every direction

- The TI envelope depends on the projection direction of E_1 and E_2.
- Field alignment influences the envelope modulation amplitude.

Electrode Placement Alters Field Direction



- The present study evaluated the TI envelope based on electric-field magnitude without explicitly considering its directional dependence.
- However, TI modulation depends on the vector relationship between the two interfering electric fields.

Future Work

Incorporate direction-dependent TI envelope analysis to better characterize bladder-targeted stimulation.

2. Grossman N, et al. Noninvasive Deep Brain Stimulation via Temporally Interfering Electric Fields. Cell. 2017;169(6):1029–1041.e16. doi:10.1016/j.cell.2017.05.024.

Conclusion

1. Phantom experiments validated the COMSOL simulation through comparable electric-field distributions.
2. 2D models revealed placement-dependent field behavior and guided electrode placement for 3D modeling.
3. 3D optimization identified Case 4 as the most selective configuration for bladder-targeted TIS.
4. Electrode geometries showed comparable performance, with the square electrode providing the highest numerical selectivity.
5. These findings provide a foundation for future experimental validation and development of bladder-targeted TIS.

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Thank You